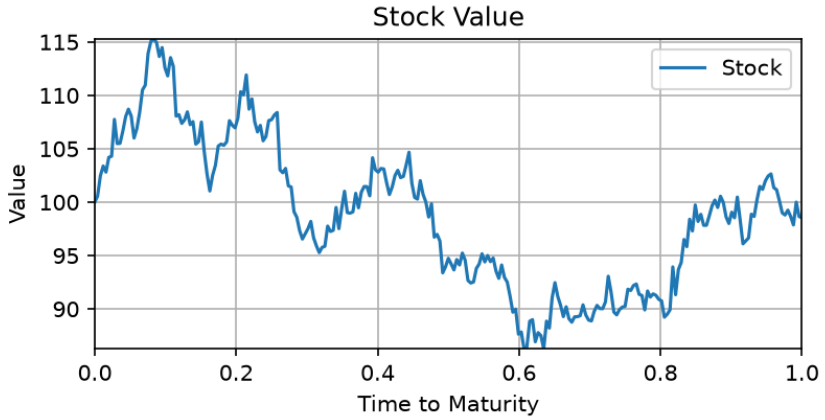
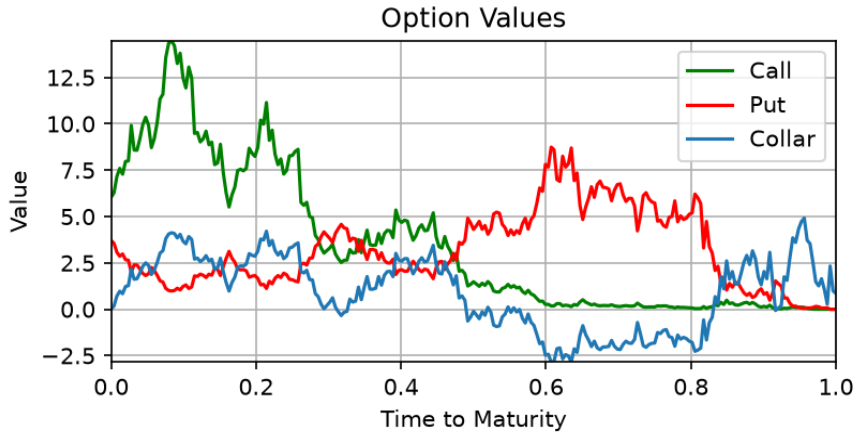
STRUCTURED EQUITY DERIVATIVES – EQUITY COLLAR

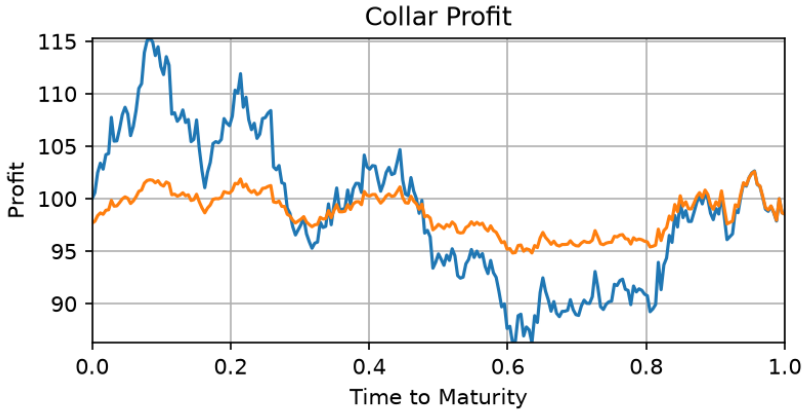
1. Generated underlying asset price path, modelled as a Geometric Brownian Motion.



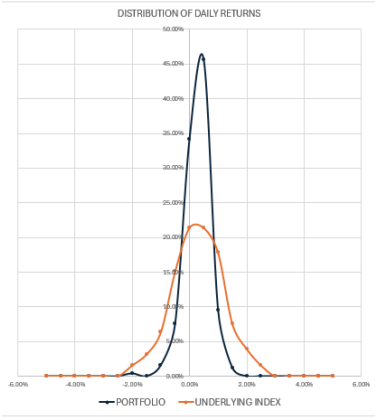
2. The daily returns to both the Put and Call option positions, and their combined running profit.

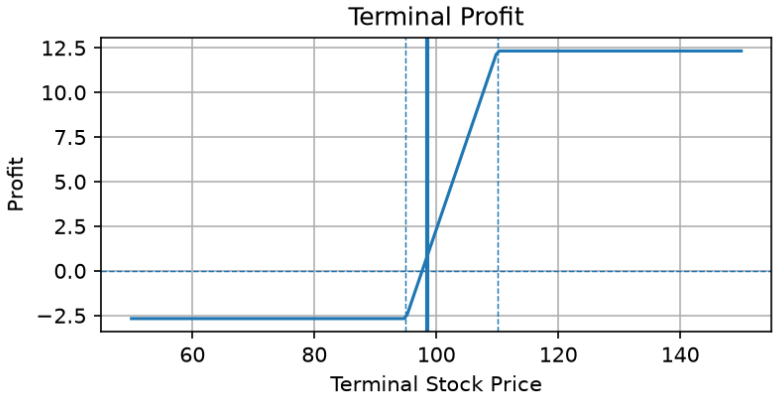


3. Compared the underlying stock value to the collar profit plus the underlying stock value.



4. Simulated 50 paths to examine the distribution of daily returns of both strategies and the payoff at maturity.





STRUCTURED EQUITY DERIVATIVES – EQUITY COLLAR

Key takeaways:

PRO's:

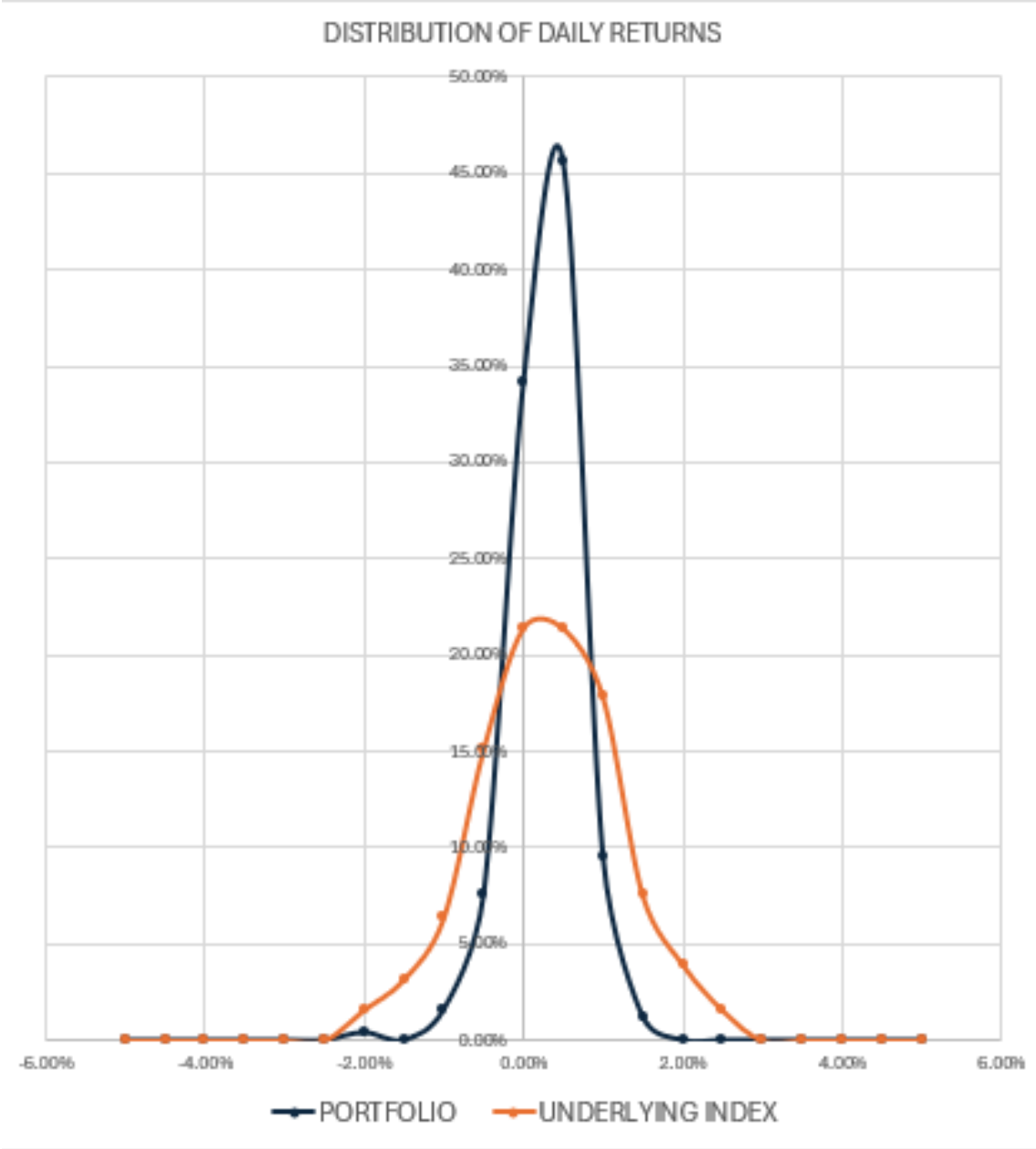
- Downside protection through the Put option – if the stock were to fall dramatically you can play out the right to sell the stock at the strike price.
- Lower cost than only buying a put as you receive the premium from the short call, which can offset some or potentially all the cost from buying the protective put. This depends on the net premium: the short call minus the long put.
- Reduces volatility (as seen by the narrower daily return distribution) to reduce portfolio risk.

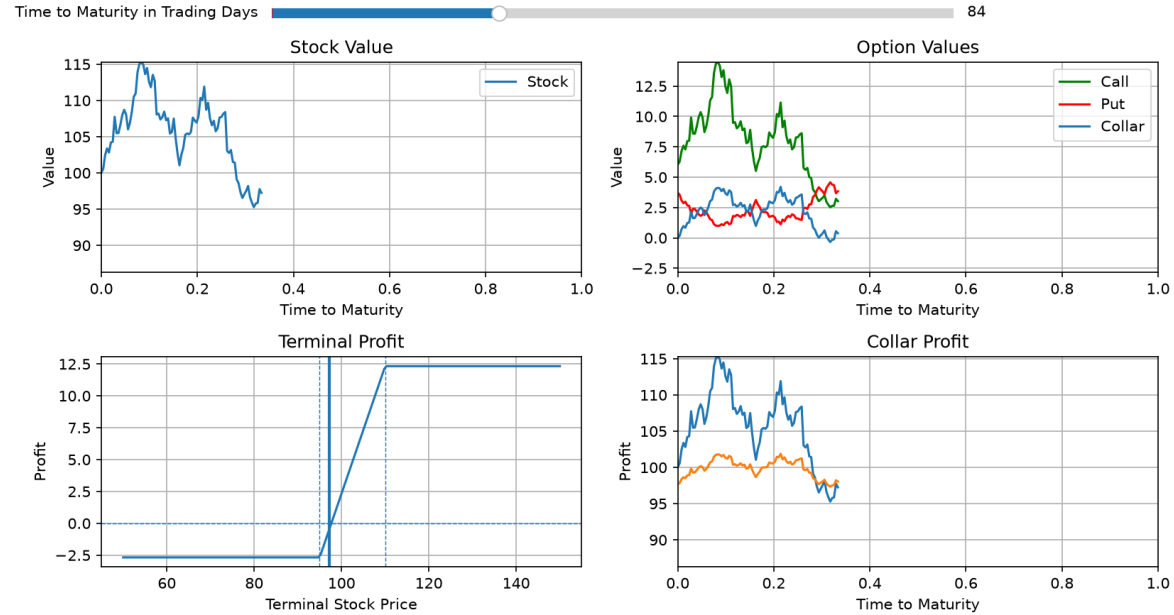
CON's:

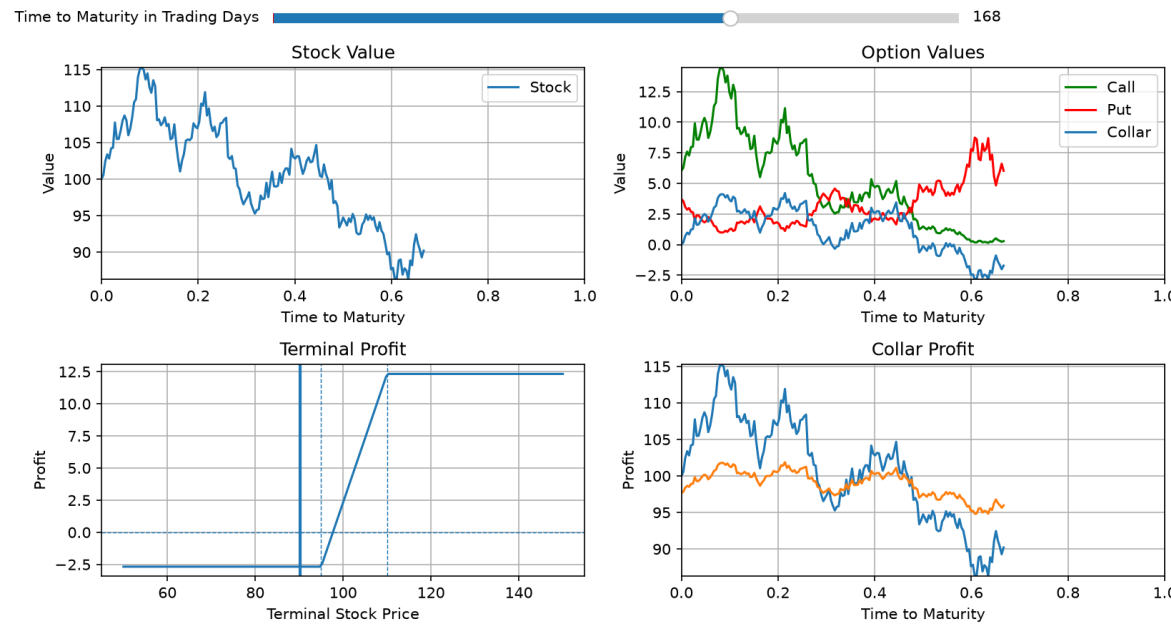
- Upside is capped – the short call creates the obligation to sell at the strike price if the stock's value exceeds the strike price.
- Path dependency: looking at simulations that have a stock price that terminates valued around 100, it can be seen that the payoff at maturity is similar, however their interim profits varied widely.
- The Collar does not eliminate losses as even if you sell the asset after it has fallen below the long-put strike price, you will still lose the difference between the initial stock price and the put strike price per share.

FUTURE WORK:

- Test a cross-hedge: sell expensive calls on one asset to buy cheaper puts on another asset, with both assets being closely correlated.
- Optimal strategy for roll-overs: wait for maturity or close and re-open options before maturity?


DISTRIBUTION OF DAILY RETURNS
50.00%
45.00%
40.00%
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
-6.00%
-4.00%
-2.00%
0.00%
2.00%
4.00%
6.00%
PORTFOLIO
UNDERLYING INDEX


Time to Maturity in Trading Days
84
Stock Value
Stock
Value
Time to Maturity
Option Values
Call
Put
Collar
Value
Time to Maturity
Terminal Profit
Profit
Terminal Stock Price
Collar Profit
Profit
Time to Maturity


Time to Maturity in Trading Days
168
Stock Value
Stock
Value
Time to Maturity
Option Values
Call
Put
Collar
Value
Time to Maturity
Terminal Profit
Profit
Terminal Stock Price
Collar Profit
Profit
Time to Maturity

Time to Maturity in Trading Days 252

